

Julian Garcia-Abadillo, PhD

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Research Summary

I apply statistical and machine learning methods to agricultural problems, mainly in plant breeding: predicting phenotypes from multi-omics data (genomic, pedigree, weather, phenomic), identifying genotype-by-environment patterns, and selecting optimal information sources for training better prediction models.

Education

- **PhD. Quantitative Genetics** — Technical University of Madrid (UPM) 09/2021 – 12/2025
PhD Advisor: Dr. Julio Isidro y Sánchez
- **MSc. Artificial Intelligence** — Technical University of Madrid (UPM) 09/2020 – 07/2021
- **BSc. Biotechnology** — Technical University of Madrid (UPM) 09/2016 – 07/2020

Work Experience

- **Research Assistant** — University of Florida, Gainesville, FL, US 09/2024 – 02/2025 | 08/2025 – present
Provide quantitative genetics and statistical support across plant breeding and tree orchard research projects; advise MS/PhD students; serve as TA and guest lecturer.
- **AI/Plant Biology PhD Resident** — Google LLC, Mountain View, CA, US 01/2024 – 07/2024
Led independent research using Google's proprietary data and infrastructure to deliver insights for client companies; produced and reviewed Python/R data science pipelines; advised software engineers and reported findings to leadership.

Selected Publications

1. **Garcia-Abadillo, J.**, Isidro Sánchez, J., Jarquin, D., Galeano, E., & Thomas, B. R. (in press). Prediction of conelet mortality rates in interior lodgepole pine using phenomic, nutritional, and environmental data. *Frontiers in Plant Science*, 17, 1845181.
2. **Garcia-Abadillo, J.**, Barba, P., Carvalho, T., Sosa-Zuniga, V., Lozano, R., Carvalho, H., Garcia-Rojas, M., Salazar, E., & Sanchez, J. (2024). Dissecting the complex genetic basis of pre-and post-harvest traits in *Vitis vinifera* L. using genome-wide association studies. *Horticulture Research*, 11(2), uhad283.
3. **Garcia-Abadillo, J.**, Adunola, P., Aguilar, F., Trujillo-Montenegro, J., Riascos, J., Persa, R., Sanchez, J., & Jarquin, D. (2024). Sparse testing designs for optimizing predictive ability in sugarcane populations. *Frontiers in Plant Science*, 15, 1400000.
4. **Garcia-Abadillo, J.**, Morales, L., Buerstmayr, H., Michel, S., Lillemo, M., Holzapfel, J., Hartl, L., Akdemir, D., Carvalho, H., & Isidro-Sanchez, J. (2023). Alternative scoring methods of fusarium head blight resistance for genomic assisted breeding. *Frontiers in Plant Science*, 13, 1057914.
5. Messina, C., **Garcia-Abadillo, J.**, Powell, O., Tomura, S., Zare, A., Ganapathysubramanian, B., & Cooper, M. (2025). Toward a general framework for AI-enabled prediction in crop improvement. *Theoretical and Applied Genetics*, 138(7), 1–15.

Selected Talks

1. **Garcia-Abadillo, J.** (2024). *Integration of multi-omics and other interactions in predictive models*. VIII International Meeting on Plant Breeding: AI for Scientific Breakthrough: Applications and Innovations, Piracicaba, Brazil.
2. **Garcia-Abadillo, J.** (2025). *Integration of multi-omics for phenotypic prediction in plant breeding*. XXXII Plant and Animal Genome Conference, San Diego, CA, US.

Software Development

- **CHiDO** (R Shiny app) — Author & Maintainer Demo | Docs
- **bayesammi** (CRAN package) — Contributor Docs

Computer Skills

- **DevOps & Version Control:** Git, GitHub
- **Computer Programming:** UNIX shell scripting, R, Python, SQL
- **Parallel Computing:** HTC/HPC platforms (HiPerGator-UF, Kairos-UPM), slurm
- **Cloud Computing:** Google Cloud Platform (Computing & Storage), VertexAI
- **Operating Systems:** Windows, macOS, Linux
- **Data Science:** scikit-learn, pandas, numpy, seaborn, tidyverse, dplyr, ggplot2
- **Web Development:** HTML, CSS, Django, Shiny
- **Desktop Editing:** Microsoft Office, LaTeX, Inkscape